

Field Visit Update Report

1. Introduction

The original project proposal focused on replacing the existing heat exchanger system with a finned tube heat exchanger to improve the efficiency of the heat transfer process used for factory air heating. The proposed concept relied on direct heat transfer from the furnace flue gas to the airflow through finned tubes, with the expectation of reducing thermal losses and enhancing overall system performance.

However, following a comprehensive field visit and consultations with factory engineers, several practical and safety-related limitations were identified, necessitating a revision of the project scope.

2. Existing System Description

The current setup operates as an indirect heat transfer system integrated with the existing factory layout and operating conditions. The process occurs in two primary stages:

- Stage 1: The boiler generates heat and elevates the temperature of the water inside the system.
- Stage 2: The hot water flows through the heat exchanger, transferring heat to the airflow used in the drying process.

In this arrangement, heat transfer occurs indirectly from the boiler to the air, utilizing hot water as the intermediate fluid.

3. Original Proposed Modification

The initial proposal aimed to replace the existing indirect heating system with a direct heat transfer finned tube heat exchanger system.

The intended concept:

- Transfer heat directly from the hot furnace flue gas to the airflow via finned tubes.
- Eliminate the intermediate water heating stage.
- Reduce heat transfer losses associated with the water loop.

Expected advantages:

- Higher overall heat transfer efficiency.
- Reduced energy losses.
- Faster heating response times.
- Lower thermal resistance due to direct heat exchange.

4. Findings from the Field Visit

After inspecting the factory infrastructure and discussing the proposed changes with the engineering staff, two critical operational and safety issues were identified.

4.1. Distance Between Furnace and Heat Exchanger

The furnace/boiler and the heat exchanger are located at a substantial distance from one another within the factory layout. Consequently:

- Routing direct flue gas to a finned tube heat exchanger would require extensive and complex ducting systems.
- Significant structural modifications to the factory infrastructure would be required.
- System complexity and installation costs would increase substantially.

4.2. Fire and Safety Concerns

Factory engineers noted that transporting high-temperature flue gases through long ducts inside the facility poses severe fire hazards. Identified risks include:

- Potential leakage of toxic, high-temperature gases.
- Overheating of adjacent structures and equipment.
- Increased risk of ignition in the working environment.
- Challenges in safely controlling and regulating high-temperature gas flows.

Due to these safety and logistical concerns, implementing a direct finned tube heat exchanger system was deemed unsuitable for the current factory environment.

4.3 Thermal Expansion and Tea Quality Concerns

Additionally, the factory engineers explained that the high temperatures associated with direct flue gas transfer in a finned tube heat exchanger could cause thermal expansion of the metal components. Due to this thermal expansion, there is a possibility of leakage and mixing between the ambient air and the flue gas streams. Such contamination could negatively affect the quality of the tea drying process and potentially damage the final tea product quality.

5. Revised Project Direction

Based on observations and technical evaluations from the field visit, the project scope has been officially revised. Rather than replacing the existing setup, the project will now focus on the Optimization of the Existing Heat Exchanger System.

The revised project aims to enhance the performance of the current indirect heating process while keeping the existing system configuration intact. The optimization study will focus on:

- Improving the thermal efficiency of the existing heat exchanger.
- Identifying and reducing heat losses throughout the system.
- Optimizing water and airflow rates for maximum efficiency.
- Investigating potential geometric improvements to the heat exchanger hardware.
- Upgrading insulation and overall thermal management.

6. Advantages of the Revised Approach

Pivoting to an optimization strategy yields several distinct benefits for the facility:

- Maintains the existing factory layout.
- Avoids major, costly structural modifications.
- Significantly reduces implementation costs.
- Minimizes operational interruptions and downtime.
- Maintains safer operating conditions.
- Delivers practical, industry-applicable improvements.

7. Conclusion

While replacing the existing indirect heating system with a direct finned tube heat exchanger initially appeared beneficial for improving heat transfer efficiency, physical inspections revealed critical practical and safety limitations. The significant distance between the boiler/furnace and the heat exchanger, combined with the severe fire risks associated with transporting high-temperature flue gases, makes the proposed direct heat transfer system unviable for this facility.

Therefore, the project direction has been effectively revised to focus on optimizing the existing indirect heat transfer system, ensuring a safer, more cost-effective, and practical approach to improving thermal performance.